

BenchPress: Benchmarks Preserving Rankings Across Multilingual Post-Training Stages

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BenchPress

Abstract

Post-training multilingual language models require repeated decisions about recipes, each evaluated on long suites relying on redundant benchmarks or biased LLM judges. We propose BenchPress, a framework that identifies which benchmarks preserve multilingual model rankings across long-context extension and post-training stages. We provide a reliability map across math, code, and long-context tasks, enabling better and cheaper decisions during multilingual model development.

1 Introduction

Modern post-training projects iterate over dozens of recipes (Lambert et al., 2025), each evaluated on long, expensive benchmark suites. Benchmarks inform decisions that impact the final model quality, but they vary in their diagnostic value. Some are sensitive to recipe changes, others have high variance, redundancy, or weak correlation with downstream objectives. In particular, long-context extension produces effective windows shorter than the advertised ones (Hsieh et al., 2024), and the later post-training stages often score open-ended tasks with biased LLM judges (Dubois et al., 2024; Feuer et al., 2025). Moreover, these reliability gaps widen for multilingual settings (Kim et al., 2025; Hada et al., 2024; Watts et al., 2024).

Heineman et al. (2025) introduce a signal-to-noise ratio (SNR) framework that quantifies benchmark reliability for English pretraining. Higher SNR benchmarks preserve model rankings across scales and reduce error in scaling-law extrapolation. Other work targets evaluation efficiency, identifying benchmark subsets with high discriminative power (Chen et al., 2025; Gupta et al., 2025; Zhou et al., 2026). All focus on English pretraining.

2 Approach

We present BenchPress, a signal-to-noise framework for post-training. Extending the definitions

from Heineman et al. (2025), for each benchmark, signal measures the variance of mean scores across recipes, noise the variance across checkpoints and seeds for a fixed recipe. We test multiple evaluation configurations, including in-language and English prompts, direct-answer and chain-of-thought (Wei et al., 2022) formats, and strict and robust extraction. We propose two decision accuracy (DA) variants. Cross-context DA measures whether short-context scores preserve recipe rankings at long context. Cross-stage DA measures whether checkpoint scores preserve rankings across post-training stages (SFT, DPO, RLVR). We analyze the correlation between SNR and each DA variant per language and per stage. We perform this analysis at the benchmark and sample level.

The model suite includes open-source multilingual models (Apertus, OLMo3, and SmolLM3) with intermediate checkpoints across post-training stages and sizes from 0.6B to 8B (Apertus et al., 2025; Olmo et al., 2025; Bakouch et al., 2025).

The benchmark suite includes reasoning, code, instruction following, regional knowledge and long context (Zhang et al., 2025; Shi et al., 2022; Dusolle et al., 2025; Singh et al., 2024; Romanou et al., 2024; Lin et al., 2022). We focus on 10 languages from 8 families (en, zh, ar, es, ja, ko, th, fr, pt, vi) and validate conclusions on additional languages.

3 Project Timeline

We first implement the signal-aware framework, optimizing the evaluation. We then launch experiments on the cluster and present preliminary results on May 24. After the milestone, we refine the definitions and run additional experiments, expanding the analysis to benchmark capability and language family. We visualize key insights and write the final report to submit on June 7 with the code.

PA will focus on evaluation setups, KP on coding, RS on mathematical reasoning, and MG on multilingual benchmarks.

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